Minute Bars from Tick Data

Bar Construction and Standard Metrics

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 4

1 What Is a Bar?

A **bar** is a summary record that compresses a sequence of raw trades (ticks) into a fixed set of statistics over a defined interval. The **minute bar** is the most common unit in intraday algorithmic trading: it aggregates all trades that occurred within a single calendar minute into a small, structured record.

A minute bar answers: *where did the price start, where did it go, and how much changed hands during this minute?*

Bars serve several purposes:

- Reduce the dimensionality of tick data (millions of events → hundreds of bars).
- Provide a regular time grid suitable for time-series models and vectorized computation.
- Expose standard metrics (OHLCV) that are inputs to virtually every intraday signal.

2 Standard OHLCV Metrics

Every bar contains five canonical fields. Trade i in the minute has price p_i and size s_i (shares).

Field	Symbol	Definition
Open	O_t	Price of the **first** trade in the minute
High	H_t	**Maximum** trade price in the minute
Low	L_t	**Minimum** trade price in the minute
Close	C_t	Price of the **last** trade in the minute
Volume	V_t	Total **shares** traded in the minute: $V_t = \sum_i s_i$

3 Extended Metrics

Beyond OHLCV, several additional statistics are routinely computed per bar.

3.1 VWAP — Volume-Weighted Average Price

The **VWAP** is the average price paid per share within the bar, weighted by trade size:

$$\mathrm{VWAP}_t = \frac{\sum_i p_i\, s_i}{\sum_i s_i} = \frac{\mathrm{DV}_t}{V_t}$$

VWAP is the standard benchmark for institutional execution quality: a buy order executed below VWAP is considered favorable; one executed above is costly. Accumulating the sums from the open gives the **session VWAP**, the benchmark most execution algorithms target.

3.2 Dollar Volume

$$\mathrm{DV}_t = \sum_i p_i\, s_i$$

Dollar volume (notional value traded) is preferred over share volume for cross-sectional comparisons, since it normalizes out differences in share price across stocks.

3.3 Trade Count

$$N_t = \text{number of individual trades in the minute}$$

Trade count is an independent proxy for activity and liquidity. High N_t with low V_t signals fragmented, small-lot trading (often algorithmic); low N_t with high V_t signals large block trades (often institutional).

3.4 Shares at Bid and Ask

Each trade can be classified as **buyer-initiated** (executed at the ask) or **seller-initiated** (executed at the bid). With $\delta_i = +1$ for a buyer-initiated trade and -1 for a seller-initiated one,

$$n_t^{\text{ask}} = \sum_{i:\,\delta_i=+1} s_i, \qquad n_t^{\text{bid}} = \sum_{i:\,\delta_i=-1} s_i.$$

The direction comes from the order book: a trade against a resting sell order executes at the ask and is buyer-initiated, and vice versa. When the book does not reveal it, the *tick rule* assigns $+1$ to an up-tick and -1 to a down-tick.

The **order flow imbalance** per bar is

$$\text{OFI}_t = n_t^{\text{ask}} - n_t^{\text{bid}}, \qquad \widetilde{\text{OFI}}_t = \frac{n_t^{\text{ask}} - n_t^{\text{bid}}}{V_t} \in [-1, 1]$$

A persistently positive OFI indicates buying pressure; negative indicates selling pressure. The normalized version $\widetilde{\text{OFI}}_t$ is comparable across minutes and stocks with very different volumes. OFI is one of the most predictive short-horizon signals in the microstructure literature.